

Diamond chips, from wafer to qubit

ADAMAS: an open, tested framework for diamond electronics and room-temperature quantum processors

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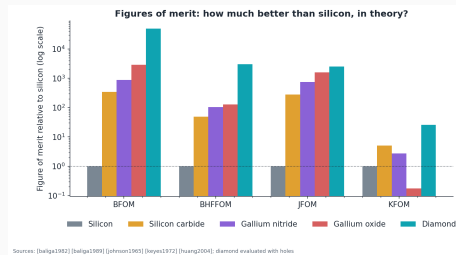
Every number on these slides is generated by the package `(make talk)`

California State University, Dominguez Hills · github.com/Normansrule/adamas-diamond-framework

Why diamond at all?

- Power-switch figure of merit: $48,976\times$ silicon
- Heat removal: $14.7\times$ silicon
- A defect, the nitrogen-vacancy (NV) center, is a **qubit at room temperature**

But: boron dopants are 0.37 eV deep (under 1% active at 300 K), wafers only reached 3 inches recently, and logic is p-channel only.



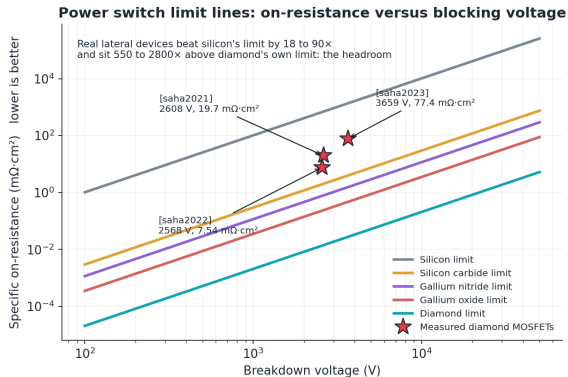
What ADAMAS is

One connected, executable chain, each link checked against published data or an independent tool:

materials → process → circuits → qubits → error correction → machine size → experiments

- 537 references, 500 verified against Crossref
- 50 figures regenerated from code
- Validation matrix: 18 checks, 17 pass
- Process traveler, PDK, placed 4-bit processor
- Circuit-level error correction (Stim, PyMatching)
- Experiments from US\$100, course, 22 web pages

Power switches: real diamond already beats silicon's limit

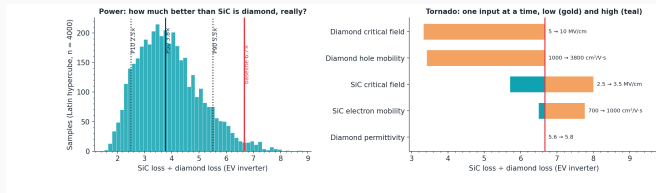


Sources: [baliga1982] [donato2020] [saha2021] [saha2022] [saha2023] [kasu2022talk]

Best measured: 3659 V; 2022 device about **89×** below silicon's limit.

800 V EV inverter	W/switch
Silicon	209.5
SiC	29.7
GaN	20.6
Diamond (ideal)	3.8

But how sure is that? Uncertainty analysis

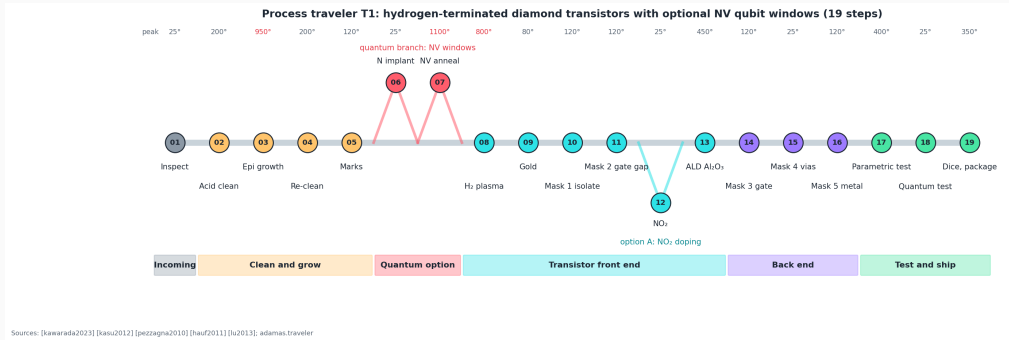


Diamond's critical field (5 to 10 MV/cm) and hole mobility (1,000 to 3,800) are not settled.

SiC ÷ diamond loss: median **3.8×**, P10 to P90 2.5 to 5.5×

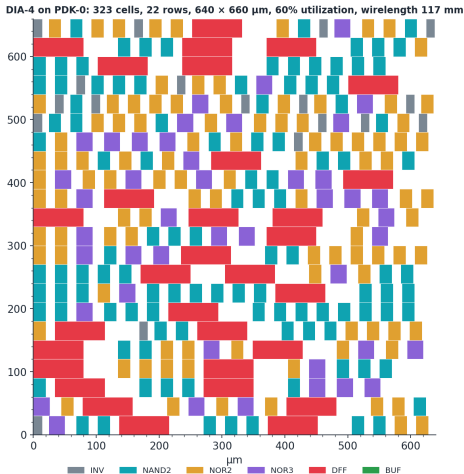
Still a big win, but the textbook value is the optimistic end.

How you would actually build it



The channel is made by surface chemistry (hydrogen termination), gold protects it, one oxygen plasma isolates every device, and **all heat is spent before the first metal.**

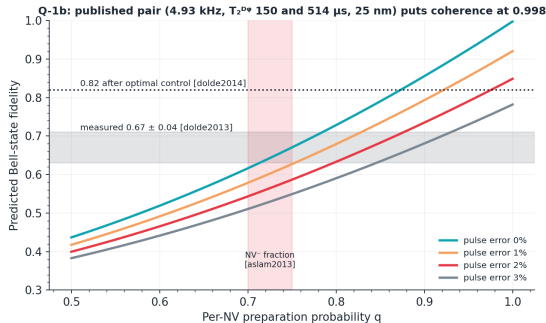
A processor on diamond



- PDK-0: layers, rules, GDS monitor die, standard cells, LEF, timed Liberty
- DIA-4 (4-bit, CALL/RET) synthesizes to 323 cells, about 1 mm² with pads
- Browser emulator matches the Verilog on random programs; ring oscillator within 10% of SPICE

Sources: [weste2011] [meadconway1980] [liu2017] [faggin1996] [ajayi2019]; circuits/digital/place.py

The two-qubit bottleneck is not coherence



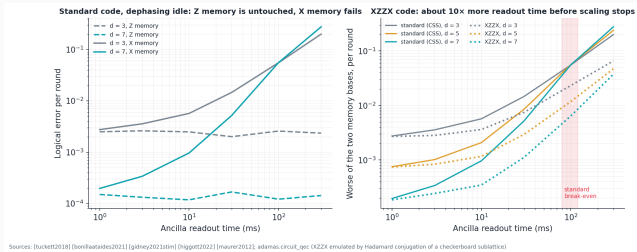
Sources: [dolde2013] [dolde2014] [aslam2013]; adamas.gate_budget model

Published room-temperature pair (25 nm, 25 μs gate): coherence alone allows **0.998**.

Measured 0.67 $\Rightarrow$ preparation $q = 0.75$;
optimal-control 0.82 $\Rightarrow q = 0.87$.

The NV^- charge state, not T_2 , limits the gate.

Error correction when readout is slow



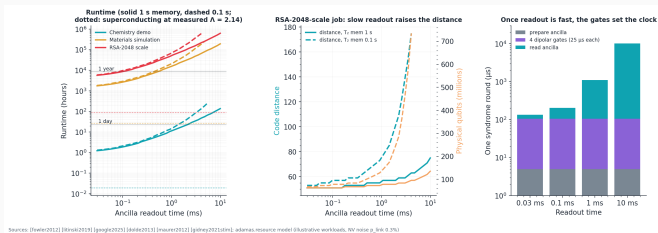
Readout takes about 1 ms while data qubits idle.

$\Lambda = 4.03$ (0.1 ms), 3.56 (1 ms), 2.64 (10 ms).

Scaling stops near 100 ms; the bias-tailored **XZZX** code pushes that to about 1000 ms.

Native circuits with biased gates:
threshold 0.95% $\rightarrow$ **1.59%** at $\eta = 100$ (standard code: 0.63%).

How big, how slow

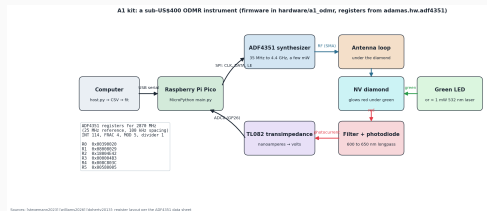
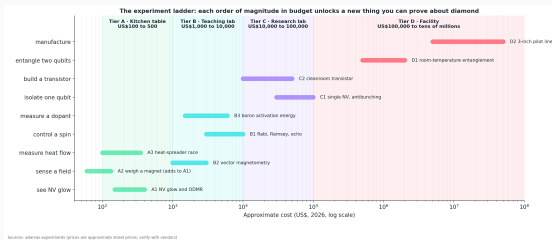


RSA-2048 scale: distance 57,
78 M qubits on a 5.7 mm die.

Runtime 6.0 years at 1 ms readout
(1.0 at 0.1 ms), P10 to P90 1.2 to
15.5 years; superconducting: 4
days.

Area is free; time is not.

Anyone can start: experiments from US\$100



A1 kit: Pico firmware drives an ADF4351 through the 2.87 GHz resonance; data go straight into the fitter.

Takeaways

1. Diamond power switches are a real advantage: **about $3.8\times$** lower loss than SiC once uncertainty is honest.
2. Room-temperature NV qubits are limited by **charge-state preparation**, not coherence.
3. A room-temperature quantum computer would be **small but slow**; readout speed and bias-tailored codes are the levers.
4. Everything is open, tested, and regenerable: `pytest`, `make figures paper talk`.

github.com/Normansrule/adamas-diamond-framework
normansrule.github.io/adamas-diamond-framework